Nonuniqueness for a convex BSDE with one exponential moment

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Abstract

We construct a finite convex generator $0 \le g(z) \le z^2/2$ and a terminal value ξ with $\mathbb{E}e^{|\xi|} < \infty$ for which the scalar BSDE has a continuum of distinct solutions. Each constructed solution satisfies the required exponential class-(D) condition and has a square-integrable integrand. This gives a negative answer to Problem 5.4 of Fan, Hu and Tang.

A process is of class (D) if its values at stopping times bounded by the terminal time form a uniformly integrable family. Fan, Hu and Tang [1, Problem 5.4, equation (5.3)] ask whether convexity, the bound $0 \le g(z) \le |z|^2/2$, and one exponential moment ensure uniqueness among solutions for which $e^{|Y|}$ is of class (D). The following construction satisfies these conditions in one Brownian dimension.

Theorem 1. *For every $T > 0$, there exist a probability measure $\mathbb{P}$, a standard Brownian motion B with its usual augmented natural filtration, a deterministic finite convex function $g : \mathbb{R} \to \mathbb{R}$ with $0 \le g(z) \le z^2/2$, and a terminal value ξ satisfying $\mathbb{E}_{\mathbb{P}} e^{|\xi|} < \infty$, such that*

$$Y_t = \xi + \int_t^T g(Z_s)\, ds - \int_t^T Z_s\, dB_s \qquad (0 \le t \le T) \tag{1}$$

has a solution (Y^c, Z^c) for every $c \in [0,1]$, with $Y_0^c = -c$. Each $e^{|Y^c|}$ is of class (D), and $\mathbb{E}_{\mathbb{P}} \int_0^T |Z_s^c|^2 ds < \infty$.

The generator. Set

$$K = \{0\} \cup \{2^n : n = 0, 1, \ldots\}, \qquad g(z) = \sup_{p \in K} \left(pz - \frac{p^2}{2} \right). \tag{2}$$

The term $p = 0$ gives $g \ge 0$, and completing the square gives $pz - p^2/2 = z^2/2 - (p-z)^2/2 \le z^2/2$. Thus g is finite and convex, hence continuous. Its affine pieces have the following useful property:

$$g(\beta q) = (\beta - \tfrac{1}{2})q^2 \quad \text{for } q = 2^n, n \ge 0, \quad \tfrac{3}{4} \le \beta \le 1. \tag{3}$$

Indeed, every $p \in K$ satisfies either $p \le q/2$ or $p \ge q$. For $z \in [3q/4, q]$,

$$\left(qz - \frac{q^2}{2} \right) - \left(pz - \frac{p^2}{2} \right) = (q-p) \left(z - \frac{q+p}{2} \right) \ge 0.$$

So the term indexed by q realizes the supremum. This argument also covers $q = 1$; it does not require $q/2 \in K$.

A stopped Brownian clock. Work first under an auxiliary measure $\mathbb{Q}$ with a standard Brownian motion W and its usual augmented natural filtration. Define

$$t_n = T(1 - 4^{-n}), \qquad h(t) = 2^n \quad (t_n \le t < t_{n+1}), \qquad h(T) = 0,$$

and, for $t < T$, let

$$a(t) = \int_0^t h(s)^2 ds, \qquad M_t = \int_0^t h(s)\, dW_s.$$

Each interval contributes $3T/4$ to the clock, so $a(t_n) = 3Tn/4 \to \infty$. The process M is a Brownian motion under this deterministic time change. Consequently its first hitting time of -4 occurs before time T almost surely. Denote this stopping time by τ, taking $\tau = T$ on the null event of no hit, and put

$$q_t = h(t)\mathbf{1}_{\{t \le \tau\}}, \qquad A_t = \int_0^t q_s^2 ds, \qquad R_t = 1 + \frac{1}{4}\int_0^t q_s\, dW_s. \tag{4}$$

The deterministic function h and the adapted left-continuous indicator $\mathbf{1}_{\{t\le\tau\}}$ make q predictable. Almost surely, $A_T = a(\tau) < \infty$, $R \ge 0$, and $R_T = 0$; R and A are constant after τ, and q vanishes there.

The probability measure. The stochastic exponential

$$N_t = \exp\left(-\int_0^t q_s\, dW_s - \frac{1}{2}A_t\right) = \exp\left(4 - 4R_t - \frac{1}{2}A_t\right) \tag{5}$$

is a positive local martingale bounded above by e^4. It is therefore a uniformly integrable martingale on $[0, T]$, and

$$N_T = e^{4 - A_T/2} > 0, \qquad \mathbb{E}_{\mathbb{Q}} N_T = 1.$$

Define $d\mathbb{P} = N_T\, d\mathbb{Q}$. These measures are equivalent, and Girsanov's theorem makes

$$B_t = W_t + \int_0^t q_s\, ds$$

a Brownian motion under $\mathbb{P}$. Moreover, Bayes' formula gives

$$D_t := N_t^{-1} = \mathbb{E}_{\mathbb{P}}[N_T^{-1} \mid \mathcal{F}_t]. \tag{6}$$

Thus D is a uniformly integrable $\mathbb{P}$-martingale. The energy is integrable under this measure:

$$\mathbb{E}_{\mathbb{P}} A_T = e^4 \mathbb{E}_{\mathbb{Q}}[A_T e^{-A_T/2}] \le e^4 \sup_{x \ge 0} x e^{-x/2} = 2e^3. \tag{7}$$

We also check the filtration. For $t < T$, define the unstopped process

$$X_t = 1 + \frac{1}{4}\int_0^t h(s)\, dB_s - \frac{1}{4}a(t).$$

It agrees with R up to τ, is positive before τ, and vanishes at τ. Hence $\tau = T \wedge \inf\{t < T : X_t = 0\}$, with $\inf \varnothing = \infty$, is a stopping time of the usual natural filtration of B. Therefore q is predictable in that filtration, and $W = B - \int q\, ds$ is adapted to it. The reverse inclusion follows from the construction of B from W. Equivalence of the measures and continuity extend this equality of augmented natural filtrations to time T.

The solutions. Under $\mathbb{P}$ we have

$$dR_t = \frac{1}{4} q_t \, dB_t - \frac{1}{4} q_t^2 dt, \qquad d \log D_t = q_t \, dB_t - \frac{1}{2} q_t^2 dt.$$

For $c \in [0,1]$, set

$$Y_t^c = \log D_t - cR_t = (4-c)R_t - 4 + \frac{1}{2} A_t, \qquad Z_t^c = (1 - c/4) q_t. \tag{8}$$

Because q takes values in K, equation (3) and $g(0) = 0$ imply

$$dY_t^c = (1 - c/4) q_t \, dB_t - (1/2 - c/4) q_t^2 dt = Z_t^c \, dB_t - g(Z_t^c) dt.$$

All these continuous processes have the same terminal value

$$\xi = \frac{1}{2} A_T - 4,$$

so integration proves (1). Equation (7) gives square integrability of every Z^c and integrability of $g(Z^c)$. The processes stop before time T, so no limiting stochastic-integral identity is needed at the terminal time.

Finally, $R \geq 0$ and $0 \leq c \leq 1$ give

$$Y_t^c \geq -4, \qquad e^{Y_t^c} = D_t e^{-cR_t} \leq D_t, \qquad e^{|Y_t^c|} \leq D_t + e^4.$$

The last inequality holds at every stopping time. By (6), its right-hand side forms a uniformly integrable family. Hence $e^{|Y^c|}$ is of class (D), and $\mathbb{E}_{\mathbb{P}} e^{|\xi|} \leq 1 + e^4$. Since $Y_0^c = -c$, the solutions are distinct. This proves Theorem 1. □

Scope. The generator is deterministic, finite and convex; differentiability and strong convexity are not among the question's hypotheses. The construction therefore refutes the universal uniqueness assertion. Higher Brownian dimensions can be treated by using only the first coordinate and replacing $g(z)$ by $g(z_1)$.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] S. Fan, Y. Hu and S. Tang. *1D nonlinear backward stochastic differential equations: a unified theory and applications*. arXiv:2309.06233v2, 11 February 2026, §1 and Problem 5.4. `https://arxiv.org/html/2309.06233v2`.